

School of Natural Sciences and Mathematics

Physics (BA, BS)

The science of physics seeks understanding of the behavior of matter and energy at the most general and fundamental level. The physicist is trained to explore the physical universe in which people live and seeks interpretations of the natural phenomena found there. While much is known about the physical universe, many phenomena still remain to be investigated, understood, and exploited to the ultimate benefit of humankind. This is the challenge that a modern physicist faces.

The Degrees

The student majoring in Physics must meet the general University requirements for admission and for the specific degree the student is seeking. The Physics Program offers both the Bachelor of Arts and the Bachelor of Science degrees. A total of 120 semester credit hours is required for either degree. With the proper sequencing of courses, these degrees can be achieved in a four year period.

Bachelor of Arts

The [Bachelor of Arts](#) program provides an opportunity for a strong base in physics for students wishing to pursue graduate studies (non-physics) in, for example, business administration, economics, finance, oceanography, and patent or high technology law. Additionally, students seeking certification as high school teachers with physics as a major specialization and those seeking employment in industry, government service, and computer technology have the opportunity to obtain the necessary physics background through the BA program. The lower-division course requirements for the BA degree are the same as those for the BS degree. At the upper-division level, the physics core course are identical with the BS core courses, but the physics elective is replaced with [PHYS 4398](#) Senior Research Project for BA Degree and 15 semester credit hours of advanced physics courses are replaced with 15 semester credit hours of upper-division science electives.

Bachelor of Science

The [Bachelor of Science](#) is intended for students interested in a professional career in physics or closely related fields. It provides an excellent background for graduate programs in physics, biophysics, geophysics, engineering, medicine and other health related degree programs.

Graduate Studies Track

The recommended course of study toward a Bachelor of Science degree for those students who intend to pursue graduate studies in Physics begins with a two-semester Honors sequence of fundamentals of physics that gives the student a more extensive foundation in basic physics. The remainder of the program is the same as the regular BS program.

Algebra Based Physics

An algebra based general physics course ([PHYS 1301](#), [PHYS 1302](#)) with lab ([PHYS 2125](#), [PHYS 2126](#)) is offered for students interested in the health sciences and those curious about the physical world in which we live. It stresses understanding the workings of nature and the physical processes and phenomena occurring therein.

Bachelor of Arts in Physics

[Degree Requirements](#) (120 semester credit hours)¹

[View an Example of Degree Requirements by Semester](#)

Faculty

FACG> nsm-physics-ba

Professors: Phillip C. Anderson, Lunjin Chen, Gerardo Cisneros, Shengwang Du, Robert Glosser, Matthew J. Goeckner, Mustapha Ishak-Boushaki, David J. Lary, Mark Lee, Xinchou Lou, Roger Malina, Anton V. Malko, Robert M. Wallace, Anvar A. Zakhidov, Chuanwei Zhang, Fan Zhang

Associate Professors: Fabiano Da Silveira Rodrigues, Yuri Gartstein, Michael Kesden, Lindsay J. King, Lloyd Lumata, Bing Lv, Jason D. Slinker, Xiaojia Zhang

Assistant Professors: Michael Kolodrubetz, Kaloyan Penev, Xiaoyan Shi, Aaron Smith, Qingyu Zhu

Professors Emeriti: Austin J. Cunningham, Walter Heikkila, Joseph M. Izen, Myron B. Salamon, Brian A. Tinsley

Research Professor: Roderick A. Heelis

Professor of Instruction: Paul Mac Alevey

Assistant Professors of Instruction: Amena Khan, Lamya Saleh

UT Dallas Affiliated Faculty: Kyeongjae (KJ) Cho, John P. Ferraris, Massimo V. Fischetti, Heather

Hayenga, Julia W. P. Hsu, Stephen D. Levene, Lawrence J. Overzet, A. Dean Sherry, Mary L. Urquhart

I. Core Curriculum Requirements: 42 semester credit hours²

Communication: 6 semester credit hours

[COMM 1311](#) Survey of Oral and Technology-based Communication

[RHET 1302](#) Rhetoric

Or select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

Mathematics: 3 semester credit hours

[MATH 2413](#) Differential Calculus^{3,4}

or [MATH 2417](#) Calculus I^{3,4}

Or select any 3 semester credit hours from [Mathematics Core](#) courses (see advisor)

Life and Physical Sciences: 6 semester credit hours

[PHYS 2325](#) Mechanics⁵

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat^{5,6}

[PHYS 2326](#) Electromagnetism and Waves⁵

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves^{5,6}

Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor)

Language, Philosophy and Culture: 3 semester credit hours

[HUMA 1301](#) Exploration of the Humanities

Or select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

Creative Arts: 3 semester credit hours

[ARTS 1301](#) Exploration of the Arts

Or select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

American History: 6 semester credit hours

Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

Government/Political Science: 6 semester credit hours

[GOVT 2305](#) American National Government

[GOVT 2306](#) State and Local Government

Or select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

Select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

Component Area Option: 6 semester credit hours

[CHEM 1311](#) General Chemistry I⁵

or [CHEM 1315](#) Honors Freshman Chemistry I⁵

[CHEM 1312](#) General Chemistry II⁵

or [CHEM 1316](#) Honors Freshman Chemistry II⁵

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

II. Major Requirements: 66-68 semester credit hours

Major Preparatory Courses: 24-26 semester credit hours beyond core curriculum

[CHEM 1111](#) General Chemistry Laboratory I

or [CHEM 1115](#) Honors Freshman Chemistry Laboratory I

[CHEM 1112](#) General Chemistry Laboratory II

or [CHEM 1116](#) Honors Freshman Chemistry Laboratory II

[CHEM 1311](#) General Chemistry I⁵

or [CHEM 1315](#) Honors Freshman Chemistry I⁵

[CHEM 1312](#) General Chemistry II⁵

or [CHEM 1316](#) Honors Freshman Chemistry II⁵

[MATH 2413](#) Differential Calculus^{3 4}_{,-}

or [MATH 2417](#) Calculus I^{3 4}_{,-}

[MATH 2414](#) Integral Calculus⁴₋

or [MATH 2419](#) Calculus II⁴₋

[MATH 2415](#) Calculus of Several Variables⁴₋

or [MATH 3351](#) Advanced Calculus⁴₋

[MATH 2418](#) Linear Algebra

[MATH 2420](#) Differential Equations with Applications

[PHYS 1100](#) The Fun of Physics

[PHYS 2303](#) Contemporary Physics

[PHYS 2325](#) Mechanics⁵₋

and [PHYS 2125](#) Physics Laboratory I⁷₋

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat^{5 6 8}_{,-,-}

[PHYS 2326](#) Electromagnetism and Waves⁵₋

and [PHYS 2126](#) Physics Laboratory II⁷₋

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves^{5 6}_{,-}

and [PHYS 2126](#) Physics Laboratory II⁷₋

Major Core Courses: 30 semester credit hours

[PHYS 3312](#) Classical Mechanics

[PHYS 3427](#) Electronics with Laboratory

[PHYS 3330](#) Numerical Methods in Physics and Computational Techniques

[PHYS 3411](#) Theoretical Physics

[PHYS 3416](#) Electricity and Magnetism

[PHYS 4311](#) Thermodynamics and Statistical Mechanics

[PHYS 4373](#) Physical Measurements Laboratory

[PHYS 4398](#) Senior Research Project for BA Degree

Choose one Physics Elective course from the following:

[PHYS 3380](#) Astronomy

[PHYS 4301](#) Quantum Mechanics I

[PHYS 4302](#) Quantum Mechanics II

[PHYS 4340](#) Introduction to Quantum Information

[PHYS 4346](#) Quantum Physics for Engineers and Programmers

[PHYS 4352](#) Concepts of Modern Physics

[PHYS 4371](#) Solid State Physics

[PHYS 4386](#) Elementary Particle Physics

[PHYS 4392](#) Extragalactic Astrophysics

[PHYS 4V07](#) Senior Research Projects

[PHYS 4V10](#) Research Topics in Physics

[PHYS 4V11](#) Topics in Physics

Major Related Courses: 12 semester credit hours

12 semester credit hours of upper-division Science Electives

III. Elective Requirements: 10-12 semester credit hours

Both lower- and upper-division courses may count as electives.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

Bachelor of Science in Physics

[Degree Requirements](#) (120 semester credit hours)¹

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[ARTS 1301](#) Exploration of the Arts

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Select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

Component Area Option: 6 semester credit hours

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[CHEM 1312](#) General Chemistry II⁵

or [CHEM 1316](#) Honors Freshman Chemistry II⁵

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II. Major Requirements: 66-68 semester credit hours

Major Preparatory Courses: 24-26 semester credit hours beyond core curriculum

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or [CHEM 1115](#) Honors Freshman Chemistry Laboratory I

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[PHYS 1100](#) The Fun of Physics

[PHYS 2303](#) Contemporary Physics⁷

[PHYS 2325](#) Mechanics⁵

and [PHYS 2125](#) Physics Laboratory I⁷

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat^{5 6 8}

[PHYS 2326](#) Electromagnetism and Waves⁵

and [PHYS 2126](#) Physics Laboratory II⁷

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves^{5 6}

and [PHYS 2126](#) Physics Laboratory II⁷

Major Core Courses: 24 semester credit hours

[PHYS 3312](#) Classical Mechanics

[PHYS 3427](#) Electronics with Laboratory

[PHYS 3330](#) Numerical Methods in Physics and Computational Techniques

[PHYS 3411](#) Theoretical Physics

[PHYS 3416](#) Electricity and Magnetism

[PHYS 4311](#) Thermodynamics and Statistical Mechanics

[PHYS 4373](#) Physical Measurements Laboratory

Major Related Courses: 18 semester credit hours

[PHYS 4301](#) Quantum Mechanics I

[PHYS 4302](#) Quantum Mechanics II

[PHYS 4328](#) Optics

[PHYS 4352](#) Concepts of Modern Physics

Choose two Physics Elective courses from the following:

[PHYS 3380](#) Astronomy

[PHYS 4340](#) Introduction to Quantum Information

[PHYS 4371](#) Solid State Physics

[PHYS 4386](#) Elementary Particle Physics

[PHYS 4392](#) Extragalactic Astrophysics

[PHYS 4V07](#) Senior Research Projects

[PHYS 4V10](#) Research Topics in Physics

[PHYS 4V11](#) Topics in Physics

III. Elective Requirements: 10-12 semester credit hours

Both lower- and upper-division courses may count as electives.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

NOTE> Please be advised, the UTeach section below feeds in from a separate page updated by the UTeach team. Any changes made to the UTeach section below will not be retained.

UTeach Option

The [UTeach option](#) may be added to either the BA or BS degree. UTeach Dallas Option degree plans are streamlined to allow students to complete both a rigorous Bachelor of Science or Bachelor of Arts degree and all coursework for middle or high school teacher certification in four years. Teaching Option degrees require deep content knowledge combined with courses grounded in the latest research on math and science education. While most graduates go on to classroom teaching, UTeach alums are also prepared to enter graduate school and to work in a discipline related industry.

Fast Track Baccalaureate/Master's Degrees

For students interested in pursuing graduate studies in physics, the Physics Department offers an accelerated BS / MS Fast Track that involves taking graduate courses in lieu of several advanced undergraduate courses. Acceptance into the Fast Track is based on the student's attaining a GPA (grade point average) of at least 3.200 on a minimum of 30 semester credit hours of upper-division courses that include [PHYS 3411](#), [PHYS 3312](#), [PHYS 3330](#), [PHYS 3416](#), [PHYS 4301](#) and [PHYS 4311](#). Eligible students may take up to 15 semester credit hours of selected graduate courses that may be used to complete the baccalaureate degree and also satisfy requirements for the master's degree. These credits will partially satisfy the UT Dallas MS degree requirements when the student completes the BS degree. Interested students should contact their advisor during their junior year to apply to the Fast Track program.

1. Incoming freshmen must enroll and complete requirements of UNIV 1010 and the corresponding school-related freshman seminar course. Students, including transfer students, who complete their core curriculum at UT Dallas must take UNIV 2020.

2. Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.
3. One semester credit hour of Calculus is counted as Major Preparatory credit; three semester credit hours are counted in Core Curriculum.
4. Students may choose either calculus sequence MATH 2413, MATH 2414, and MATH 2415 or MATH 2417, MATH 2419 and MATH 3351.
5. A required Major preparatory course that also fulfills a Core Curriculum requirement. Semester credit hours are counted in Core Curriculum.
6. If Honors Physics is taken, three hours are counted under the sciences core and one hour is counted under Physics Major Preparatory Courses.
7. Indicates a prerequisite class to be completed before enrolling for upper-division classes.
8. Students who complete PHYS 2421 do not need to complete PHYS 2125.

Minors

Students must take a minimum of 18 semester credit hours for the minor, 12 of which must be upper-division semester credit hours. Students who take a minor will be expected to meet the normal prerequisites in courses making up the minor, and should maintain a minimum GPA of 2.000 on a 4.00 scale (C average). Semester credit hours may not be used to satisfy both the major and minor requirements; however, free elective semester credit hours or major preparatory classes may be used to satisfy the minor.

For all minors in the School of Natural Sciences and Mathematics students must complete all prerequisite sequences for required minor courses.

Minor in Physics

20 semester credit hours

Required: 12 semester credit hours

[PHYS 2325](#) Mechanics

[PHYS 2125](#) Physics Laboratory I

[PHYS 2326](#) Electromagnetism and Waves

[PHYS 2126](#) Physics Laboratory II

[PHYS 3411](#) Theoretical Physics

Also:

Three other upper-division physics courses

Certificates

Certificate in Quantum Information

15 semester credit hours

A Certificate in Quantum Information is offered by the Department of Physics in the School of Natural Sciences and Mathematics.

The focus of the Certificate in Quantum Information is to provide science oriented students with the necessary skills and knowledge for job opportunities in the quantum industry.

Admission Requirements

Undergraduate students who have completed two semesters of undergraduate calculus and one semester of undergraduate linear algebra.

Certificate Requirements

Students must complete the following courses:

[PHYS 4346](#) Quantum Physics for Engineers and Programmers

or [PHYS 4301](#) Quantum Mechanics I

[PHYS 3330](#) Numerical Methods in Physics and Computational Techniques⁹

[PHYS 4340](#) Introduction to Quantum Information

[PHYS 4350](#) Quantum Algorithm and Software

[PHYS 4347](#) Quantum Network and Communication

9. Equivalent programming courses, such as CS 1325, CS 1336, BMEN 1300, or MATH 2370, can substitute for PHYS 3330.

Updated: 2023-06-29 11:17:49 v15.4122fc